

C:\Users\Admin\OneDrive - South East Technological University\12. Paper 4\Post workshop\Element 2 \Result\Element2_PathAnalysis_Thi Minh Thuan Phung_20115516.amw

Analysis Summary

Date and Time

Date: Wednesday, July 29, 2026

Time: 4:34:52 PM

Title

Element2_pathanalysis_thi minh thuan phung_20115516: Wednesday, July 29, 2026 4:34 PM

Groups

Group number 1 (Group number 1)

Notes for Group (Group number 1)

The model is recursive.

Sample size = 100

Variable Summary (Group number 1)

Your model contains the following variables (Group number 1)

Observed, endogenous variables

ID5
ID4
ID3
ID2
ID1
OC4
OC2
OC1
SI4
SI3

SI2
 SI1
 ES4
 ES3
 ES2
 ES1
 HB4
 HB3
 HB2
 HB1

Unobserved, endogenous variables

Investment_Decision

Unobserved, exogenous variables

e1
 e2
 e3
 e4
 e5
 Overconfidence
 e6
 e7
 e8
 Social_influence
 e9
 e10
 e11
 e12
 Emotional_State
 e13
 e14
 e15
 e16
 Herding_Behaviour
 e17
 e18
 e19
 e20
 e21

Variable counts (Group number 1)

Number of variables in your model: 46
Number of observed variables: 20
Number of unobserved variables: 26
Number of exogenous variables: 25
Number of endogenous variables: 21

Parameter Summary (Group number 1)

	Weights	Covariances	Variances	Means	Intercepts	Total
Fixed	26	0	0	0	0	26
Labeled	0	0	0	0	0	0
Unlabeled	19	8	25	0	0	52
Total	45	8	25	0	0	78

Models

Default model (Default model)

Notes for Model (Default model)

Computation of degrees of freedom (Default model)

Number of distinct sample moments: 210

Number of distinct parameters to be estimated: 52

Degrees of freedom (210 - 52): 158

Result (Default model)

Minimum was achieved

Chi-square = 344.989

Degrees of freedom = 158

Probability level = .000

Group number 1 (Group number 1 - Default model)

Estimates (Group number 1 - Default model)

Scalar Estimates (Group number 1 - Default model)

Maximum Likelihood Estimates

Regression Weights: (Group number 1 - Default model)

	Estimate	S.E.	C.R.	P Label
Investment_Decision <--- Overconfidence	.470	.290	1.623	.105
Investment_Decision <--- Herding_Behaviour	.134	.125	1.073	.283
Investment_Decision <--- Emotional_State	.426	.178	2.398	.016
Investment_Decision <--- Social_influence	-.052	.205	-.255	.799

ID5	<--- Investment_Decision	1.000			
ID4	<--- Investment_Decision	1.009	.133	7.607	***
ID2	<--- Investment_Decision	.711	.113	6.309	***
ID1	<--- Investment_Decision	.631	.117	5.377	***
OC4	<--- Overconfidence	1.000			
OC2	<--- Overconfidence	1.657	.360	4.600	***
OC1	<--- Overconfidence	1.329	.293	4.540	***
SI4	<--- Social_influence	1.000			
SI3	<--- Social_influence	1.078	.136	7.934	***
SI2	<--- Social_influence	.858	.126	6.814	***
SI1	<--- Social_influence	.702	.114	6.141	***
ES4	<--- Emotional_State	1.000			
ES3	<--- Emotional_State	1.034	.132	7.837	***
ES2	<--- Emotional_State	.659	.107	6.141	***
ES1	<--- Emotional_State	.686	.119	5.743	***
HB4	<--- Herding_Behaviour	1.000			
HB3	<--- Herding_Behaviour	.966	.111	8.720	***
HB2	<--- Herding_Behaviour	1.050	.113	9.317	***
HB1	<--- Herding_Behaviour	1.133	.116	9.774	***
ID3	<--- Investment_Decision	.602	.114	5.288	***

Standardized Regression Weights: (Group number 1 - Default model)

	Estimate
Investment_Decision <--- Overconfidence	.237
Investment_Decision <--- Herding_Behaviour	.127
Investment_Decision <--- Emotional_State	.430
Investment_Decision <--- Social_influence	-.049
ID5 <--- Investment_Decision	.847
ID4 <--- Investment_Decision	.759
ID2 <--- Investment_Decision	.637
ID1 <--- Investment_Decision	.552
OC4 <--- Overconfidence	.534
OC2 <--- Overconfidence	.864
OC1 <--- Overconfidence	.673
SI4 <--- Social_influence	.792
SI3 <--- Social_influence	.818
SI2 <--- Social_influence	.703
SI1 <--- Social_influence	.642
ES4 <--- Emotional_State	.786
ES3 <--- Emotional_State	.832
ES2 <--- Emotional_State	.644
ES1 <--- Emotional_State	.606
HB4 <--- Herding_Behaviour	.811

HB3	<--- Herding_Behaviour	.795
HB2	<--- Herding_Behaviour	.836
HB1	<--- Herding_Behaviour	.871
ID3	<--- Investment_Decision	.544

Covariances: (Group number 1 - Default model)

		Estimate	S.E.	C.R.	PLabel
Overconfidence	<--> Social_influence	.139	.052	2.664	.008
Overconfidence	<--> Herding_Behaviour	.107	.047	2.278	.023
Overconfidence	<--> Emotional_State	.096	.050	1.928	.054
Social_influence	<--> Herding_Behaviour	-.073	.075	-.963	.335
Social_influence	<--> Emotional_State	.477	.109	4.394	***
Emotional_State	<--> Herding_Behaviour	-.053	.081	-.655	.512
e15	<--> e16	.127	.063	2.016	.044
e11	<--> e12	.116	.059	1.959	.050

Correlations: (Group number 1 - Default model)

	Estimate
Overconfidence <--> Social_influence	.406
Overconfidence <--> Herding_Behaviour	.311
Overconfidence <--> Emotional_State	.263
Social_influence <--> Herding_Behaviour	-.113
Social_influence <--> Emotional_State	.697
Emotional_State <--> Herding_Behaviour	-.076
e15 <--> e16	.245
e11 <--> e12	.251

Variances: (Group number 1 - Default model)

	Estimate	S.E.	C.R.	PLabel
Overconfidence	.183	.071	2.563	.010
Social_influence	.639	.145	4.396	***
Emotional_State	.732	.169	4.328	***
Herding_Behaviour	.651	.138	4.728	***
e21	.513	.118	4.362	***
e1	.283	.075	3.784	***
e2	.538	.103	5.216	***
e3	.619	.095	6.510	***
e4	.531	.086	6.167	***
e5	.651	.100	6.486	***
e6	.459	.073	6.313	***
e7	.171	.077	2.214	.027

e8	.391	.074	5.260	***
e9	.381	.077	4.928	***
e10	.368	.082	4.510	***
e11	.480	.084	5.748	***
e12	.449	.074	6.074	***
e13	.453	.092	4.902	***
e14	.347	.084	4.111	***
e15	.450	.074	6.088	***
e16	.595	.095	6.244	***
e17	.339	.062	5.511	***
e18	.355	.062	5.681	***
e19	.308	.060	5.168	***
e20	.266	.059	4.525	***

Squared Multiple Correlations: (Group number 1 - Default model)

	Estimate
Investment_Decision	.286
HB1	.758
HB2	.700
HB3	.631
HB4	.657
ES1	.367
ES2	.414
ES3	.693
ES4	.618
SI1	.412
SI2	.495
SI3	.668
SI4	.627
OC1	.453
OC2	.747
OC4	.285
ID1	.305
ID2	.406
ID3	.296
ID4	.576
ID5	.718

Modification Indices (Group number 1 - Default model)

Covariances: (Group number 1 - Default model)

--	--

		M.I. Par Change	
e16 <-->	Overconfidence	11.187	.110
e15 <-->	Herding_Behaviour	4.482	.115
e15 <-->	Overconfidence	11.162	-.096
e15 <-->	e20	4.727	-.091
e15 <-->	e18	4.215	.091
e14 <-->	e19	5.155	-.104
e11 <-->	Overconfidence	6.501	.077
e11 <-->	e21	4.788	-.127
e9 <-->	e15	7.841	.134
e9 <-->	e11	4.269	-.102
e8 <-->	e21	12.635	.198
e8 <-->	e16	8.056	.148
e8 <-->	e14	11.846	-.169
e8 <-->	e9	7.995	-.136
e7 <-->	e21	5.137	-.112
e7 <-->	e20	4.102	.076
e7 <-->	e15	4.869	-.090
e7 <-->	e14	6.058	.107
e7 <-->	e11	5.129	.096
e6 <-->	Social_influence	9.504	.147
e6 <-->	e9	8.322	.144
e5 <-->	Emotional_State	4.373	-.133
e5 <-->	Social_influence	7.266	.153
e5 <-->	Overconfidence	5.808	-.086
e5 <-->	e10	7.142	.161
e4 <-->	Herding_Behaviour	5.702	.148
e4 <-->	Overconfidence	4.866	-.072
e4 <-->	e10	6.860	.146
e4 <-->	e6	6.070	-.134
e3 <-->	e13	8.191	.180
e3 <-->	e8	4.023	.111
e3 <-->	e4	4.962	.139
e2 <-->	Herding_Behaviour	4.648	-.143
e2 <-->	e14	6.326	.149
e1 <-->	e8	6.713	.119
e1 <-->	e3	6.689	-.139

Variances: (Group number 1 - Default model)

	M.I. Par Change
--	------------------------

Regression Weights: (Group number 1 - Default model)

	M.I. Par Change
--	------------------------

HB2 <---	ID5	4.501	.134
HB4 <---	Emotional_State	5.500	-.193
HB4 <---	Social_influence	6.947	-.231
HB4 <---	Overconfidence	6.760	-.435
HB4 <---	ES3	5.792	-.147
HB4 <---	ES4	4.418	-.125
HB4 <---	SI1	9.156	-.225
HB4 <---	SI4	6.454	-.163
HB4 <---	OC2	6.267	-.198
HB4 <---	OC4	5.760	-.194
ES1 <---	Overconfidence	9.029	.597
ES1 <---	OC1	14.204	.344
ES1 <---	OC2	6.799	.245
ES2 <---	Overconfidence	6.699	-.452
ES2 <---	OC1	6.190	-.199
ES2 <---	OC2	7.774	-.230
ES3 <---	HB2	5.805	-.173
ES3 <---	OC1	6.480	-.218
SI2 <---	Overconfidence	6.325	.459
SI2 <---	OC1	5.616	.198
SI2 <---	OC2	7.570	.237
SI3 <---	ID1	5.704	.178
SI3 <---	ID2	4.771	.167
SI4 <---	ES2	5.831	.196
SI4 <---	OC1	6.126	-.209
OC1 <---	Investment_Decision	5.279	.200
OC1 <---	ES3	8.355	-.185
OC1 <---	SI4	8.012	-.191
OC1 <---	ID3	7.751	.202
OC1 <---	ID5	8.324	.196
OC2 <---	ID1	5.195	-.142
OC4 <---	Herding_Behaviour	7.045	-.245
OC4 <---	Emotional_State	4.044	.180
OC4 <---	Social_influence	11.415	.323
OC4 <---	HB1	6.383	-.170
OC4 <---	HB4	7.754	-.198
OC4 <---	SI1	8.043	.229
OC4 <---	SI3	6.969	.177
OC4 <---	SI4	16.088	.281
ID1 <---	OC2	4.023	-.205
ID2 <---	OC4	6.517	-.246
ID3 <---	Herding_Behaviour	4.468	.225
ID3 <---	Overconfidence	8.820	.626
ID3 <---	HB3	5.007	.186
ID3 <---	ES4	5.787	.180

ID3	<---	OC1	10.109	.307
ID3	<---	OC2	6.152	.247
ID4	<---	Herding_Behaviour	5.982	-.263
ID4	<---	HB1	5.343	-.182
ID4	<---	HB2	4.378	-.170
ID4	<---	HB3	5.158	-.191
ID4	<---	ES3	4.398	.163
ID5	<---	ES4	4.823	-.136
ID5	<---	OC1	5.356	.185
ID5	<---	ID3	4.591	-.154

Minimization History (Default model)

Iteration		Negative eigenvalues	Condition #	Smallest eigenvalue	Diameter	FN	Tries	Ratio
0	e	13		-.550	9999.000	1184.012	0	9999.000
1	e*	6		-.111	3.234	567.287	20	.493
2	e	2		-.018	.743	449.335	5	.780
3	e	0	791.737		.667	382.655	5	.892
4	e	0	694.709		1.141	367.387	3	.000
5	e	0	445.929		.945	365.482	1	.078
6	e	0	541.173		.339	349.730	1	1.168
7	e	0	308.970		.380	345.861	1	1.135
8	e	0	311.329		.125	345.035	1	1.114
9	e	0	296.025		.040	344.989	1	1.044
10	e	0	295.540		.003	344.989	1	1.004
11	e	0	294.064		.000	344.989	1	.999

Model Fit Summary

CMIN

Model	NPAR	CMIN	DF	P	CMIN/DF
Default model	52	344.989	158	.000	2.183
Saturated model	210	.000	0		
Independence model	20	1177.734	190	.000	6.199

RMR, GFI

Model	RMR	GFI	AGFI	PGFI
Default model	.089	.765	.688	.576
Saturated model	.000	1.000		
Independence model	.278	.366	.299	.331

Baseline Comparisons

Model	NFI Delta1	RFI rho1	IFI Delta2	TLI rho2	CFI
Default model	.707	.648	.817	.772	.811
Saturated model	1.000		1.000		1.000
Independence model	.000	.000	.000	.000	.000

Parsimony-Adjusted Measures

Model	PRATIO	PNFI	PCFI
Default model	.832	.588	.674
Saturated model	.000	.000	.000
Independence model	1.000	.000	.000

NCP

Model	NCP	LO 90	HI 90
Default model	186.989	137.163	244.556
Saturated model	.000	.000	.000
Independence model	987.734	883.187	1099.758

FMIN

Model	FMIN	F0	LO 90	HI 90
Default model	3.485	1.889	1.385	2.470
Saturated model	.000	.000	.000	.000
Independence model	11.896	9.977	8.921	11.109

RMSEA

Model	RMSEA	LO 90	HI 90	PCLOSE
Default model	.109	.094	.125	.000
Independence model	.229	.217	.242	.000

AIC

Model	AIC	BCC	BIC	CAIC
Default model	448.989	476.989	584.457	636.457
Saturated model	420.000	533.077	967.086	1177.086
Independence model	1217.734	1228.503	1269.838	1289.838

ECVI

Model	ECVI	LO 90	HI 90	MECVI
Default model	4.535	4.032	5.117	4.818
Saturated model	4.242	4.242	4.242	5.385
Independence model	12.300	11.244	13.432	12.409

HOELTER

Model	HOELTER .05	HOELTER .01
Default model	55	59
Independence model	19	21

Execution time summary

Minimization: .044
Miscellaneous: .274
Bootstrap: .000
Total: .318